

Practical 3 (References)



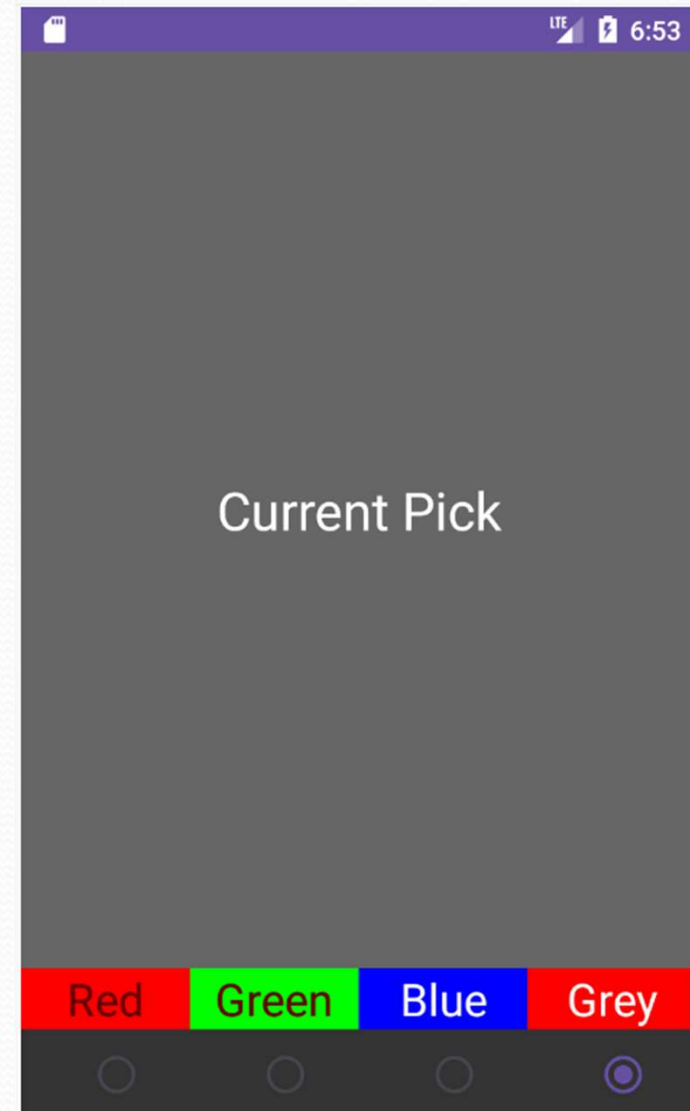


Using XML and Dynamic UI

- We learnt that one can either use XML or dynamic codes to design UI
- They could be used together, especially when a static UI is designed using XML and requires some dynamic response

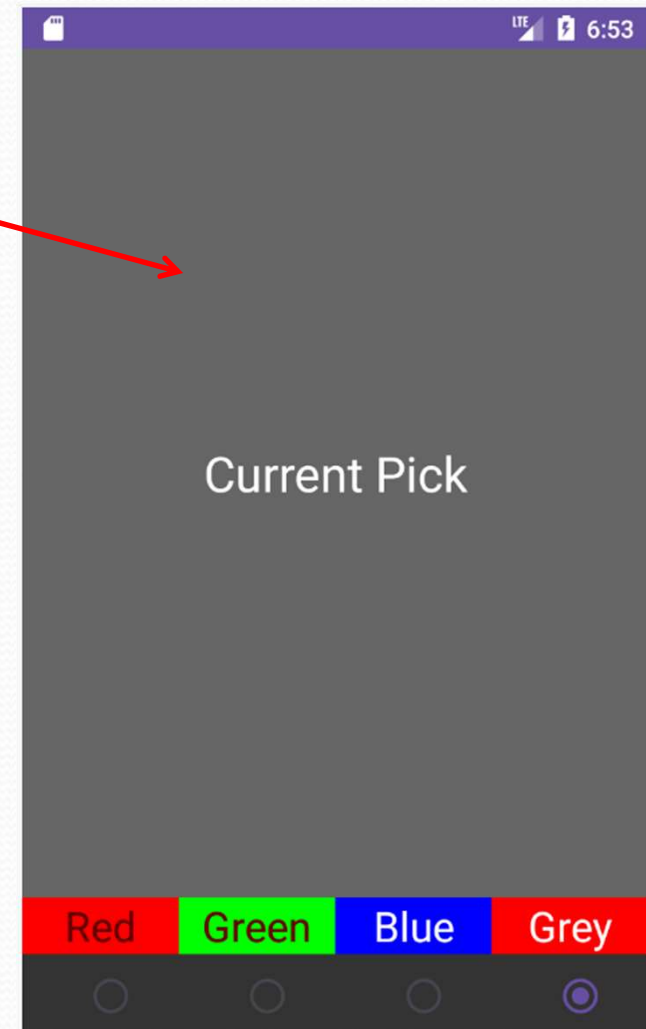
Example

- Four radio buttons that allows the color of the background to be picked
- Using XML to define the UI
- Using dynamic codes to alter the behavior



TextView

```
<TextView
    android:id="@+id/text_pick"
    android:text="@string/current_pick"
    android:textColor="#ffffff"
    android:background="#666666"
    android:layout_width="match_parent"
    android:layout_height="0dp"
    android:gravity="center"
    android:layout_weight="1"
    android:textSize="28sp">
</TextView>
```

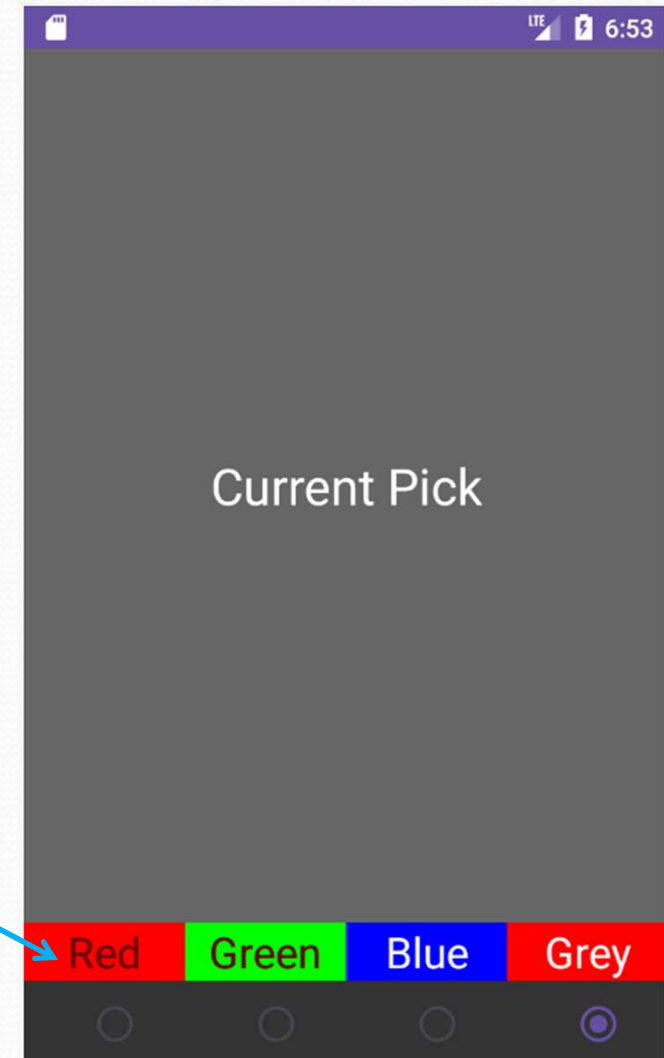


TextView

```
<LinearLayout  
    android:orientation="horizontal"  
    android:layout_width="match_parent"  
    android:layout_height="wrap_content">
```

```
    <TextView  
        android:text="@string/color_red"  
        android:textColor="#660000"  
        android:background="#ff0000"  
        android:layout_width="0dp"  
        android:layout_height="wrap_content"  
        android:layout_weight="1"  
        android:gravity="center"  
        android:textSize="24sp" />
```

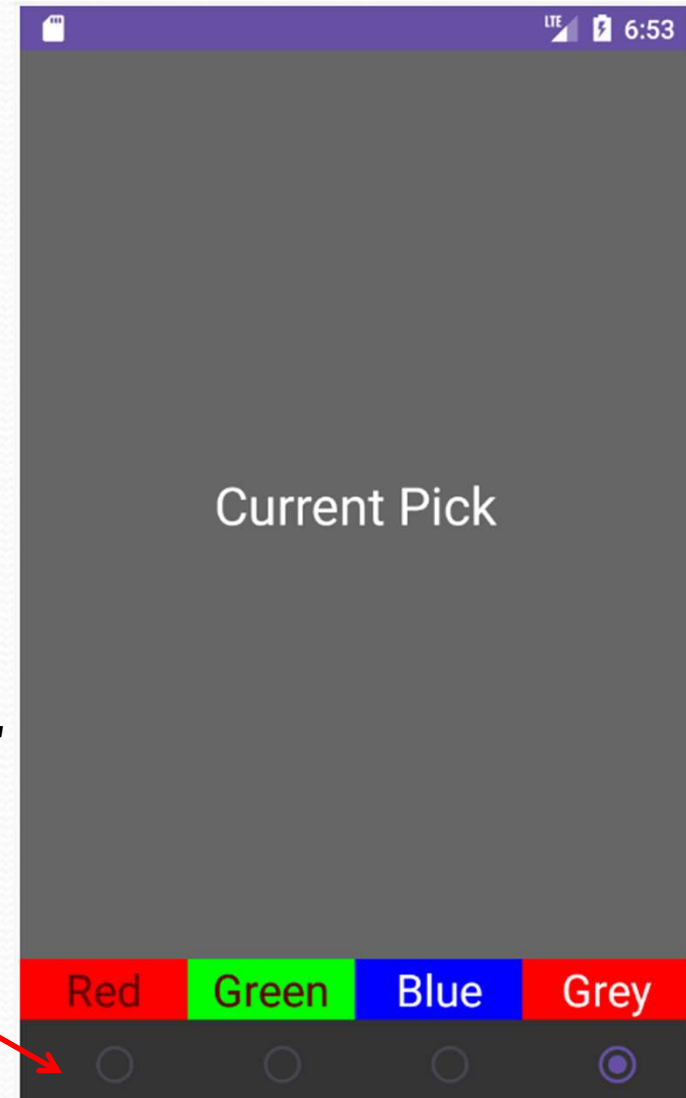
```
</LinearLayout>
```



RadioButton

```
<RadioGroup
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="horizontal"
    android:background="#333333">

    <RadioButton
        android:id="@+id/radio_grey"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:layout_weight="1"
        android:layout_marginLeft="35dp"
        android:checked="true" />
</RadioGroup>
```



Activity

- Declare variables

```
View.OnClickListener radioListener;  
TextView tvPick;
```

- Create RadioListener

```
radioListener = new View.OnClickListener() {  
    public void onClick(View v) {  
        RadioButton rb = (RadioButton)v;  
  
        if (rb.getId() == R.id.radio_red)  
            tvPick.setBackgroundColor(Color.RED);  
        else if (rb.getId() == R.id.radio_green)  
            tvPick.setBackgroundColor(Color.GREEN);  
        else if (rb.getId() == R.id.radio_blue)  
            tvPick.setBackgroundColor(Color.BLUE);  
        else if (rb.getId() == R.id.radio_grey)  
            tvPick.setBackgroundColor(Color.parseColor("#666666"));  
    }  
};
```

Activity

- Identify each components or elements

```
tvPick = findViewById(R.id.text_pick);  
RadioButton rbRed = findViewById(R.id.radio_red);  
RadioButton rbGreen =  
    findViewById(R.id.radio_green);  
RadioButton rbBlue = findViewById(R.id.radio_blue);  
RadioButton rbGrey = findViewById(R.id.radio_grey);
```

Activity

- Assign RadioListenser to each RadioButton

```
rbRed.setOnClickListener (radioListener);  
rbGreen.setOnClickListener (radioListener);  
rbBlue.setOnClickListener (radioListener);  
rbGrey.setOnClickListener (radioListener);
```

Option Menu - example

```
public boolean onCreateOptionsMenu(Menu menu) {
    super.onCreateOptionsMenu(menu);

    getMenuInflater().inflate(R.menu.main, menu);

    menu.add("Next")
        .setIntent(new Intent(this, NextActivity.class));
    menu.add("Exit")
        .setOnMenuItemClickListener(new MenuItem.OnMenuItemClickListener() {
            @Override
            public boolean onMenuItemClick(MenuItem item) {
                finish();
                return false;
            }
        });
    return super.onCreateOptionsMenu(menu);
}
```

Option Menu - example

```
public boolean onOptionsItemSelected(MenuItem item) {  
  
    int id = item.getItemId();  
  
    if (id == R.id.action_settings) {  
        //Do something here  
  
        return true;  
    }  
  
    return super.onOptionsItemSelected(item);  
}
```

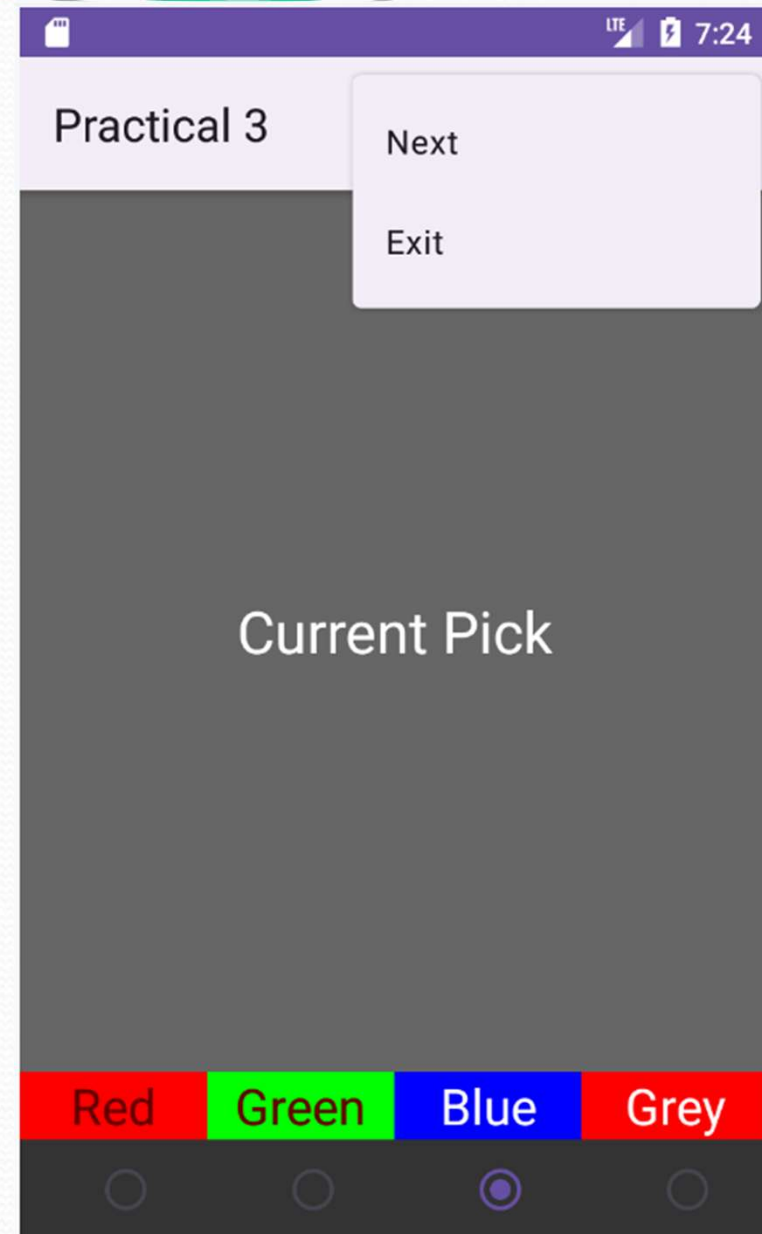
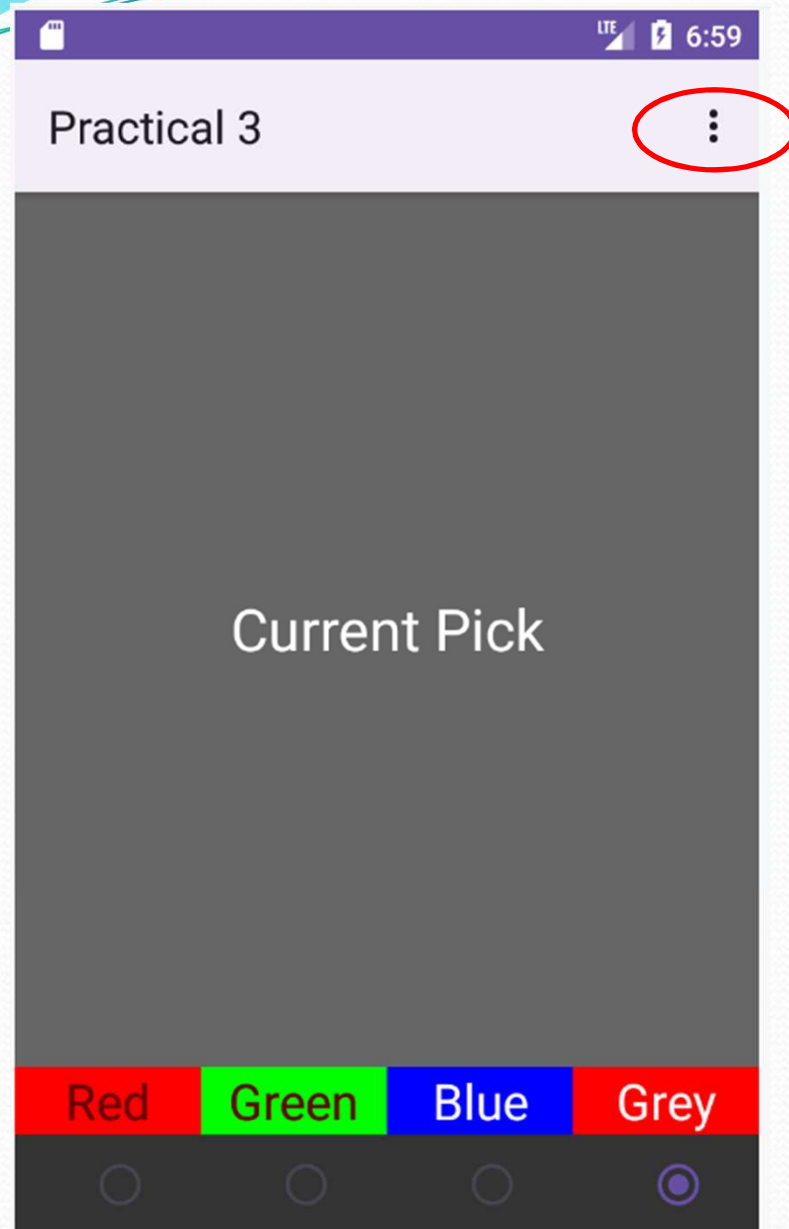
Option Menu - example

- The Action Bar by default is taken out from UI nowadays.
- To get Action Bar to appear, you need to take out this segment of code, *NoActionBar*, from themes.xml (located in res\values\themes)
- Then only the three-dot option menu can appear.

```
<style name="Base.Theme.ViewExample" parent="Theme.Material3.DayNight.NoActionBar">
```



```
<style name="Base.Theme.ViewExample" parent="Theme.Material3.DayNight">
```



Fragment - example

- Replace background TextView for display using multiple Fragments
 - Replace the TextView with FragmentContainerView

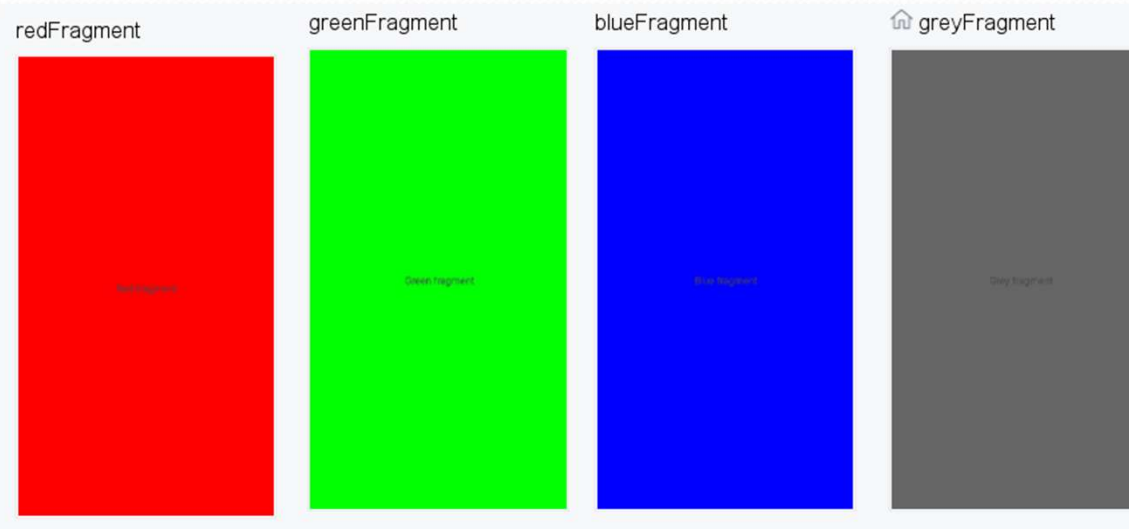
```
<androidx.fragment.app.FragmentContainerView
    android:id="@+id/nav_host_fragment"
    android:name="androidx.navigation.fragment.NavHostFragment"
    android:layout_width="match_parent"
    android:layout_height="0dp"
    android:layout_weight="1"
    app:defaultNavHost="true"
    app:navGraph="@navigation/nav_graph" />
```

- Create four Fragments, one for each colour
- Remember to add an id to each Fragment, e.g.

```
android:id="@+id/greyFragment"
```

Fragment - example

- Create a Navigation Graph (nav_graph.xml)
 - Using File->New->Android Resource File under resource folder, then select resource type “Navigation”
 - Then add new destinations, and place the Fragments one by one into nav_graph.xml



Fragment - example

- Modify MainActivity to use NavController to manage transitions to Fragment in response to RadioButton selection

- Declare variable

```
private NavController navController;
```

- Create NavHost

```
NavHostFragment navHostFragment = (NavHostFragment)  
    getSupportFragmentManager()  
        .findFragmentById(R.id.nav_host_fragment);
```

- Instantiate NavController

```
if (navHostFragment != null) {  
    navController =  
        navHostFragment.getNavController();  
}
```

Fragment - example

- Modify RadioListener implementation to navigate to the respective Fragment
 - For example, when grey RadioButton is clicked

```
navController.navigate(R.id.greyFragment);
```
- Changing to use Fragment allows separate View to be designed for each Fragment

Context Menu - example

```
public void onCreateContextMenu(ContextMenu menu, View v,  
    ContextMenuInfo menuInfo) {  
    super.onCreateContextMenu(menu, v, menuInfo);  
  
    if (v.getId() == R.id.text_pick) {  
        //getMenuInflater().inflate(R.menu.main, menu);  
        menu.add(0, 1, 0, "Easy");  
        menu.add(0, 2, 0, "Medium");  
        menu.add(0, 3, 0, "Hard");  
        menu.setHeaderIcon(android.R.drawable.ic_media_play)  
            .setTitle("Settings");  
    }  
}
```

Context Menu - example

```
public boolean onOptionsItemSelected(MenuItem item) {  
    super.onOptionsItemSelected(item);  
  
    TextView tv = findViewById(R.id.text_pick);  
  
    switch (item.getItemId()) {  
        case 1: tv.setText("You have choosen Easy"); break;  
        case 2: tv.setText("You have choosen Medium"); break;  
        case 3: tv.setText("You have choosen Hard"); break;  
    };  
  
    return true;  
}
```

Context Menu - example

- The View object that will trigger context menus must be registered

- Example:

```
registerForContextMenu (viewWithContext) ;
```

